

FINALS

Track 2

Due date: December 12, 2023.

Instructions: Please put your name and perm number on the answer sheet. You are not allowed to collaborate with other people on the finals. Remember to include the self-assessment grades on the first sheet of your finals just like a regular home work (see handout for details).

Problem 1. Let $p \in \mathbb{R}^n$ such that $p_i > 0$ for $i = 1, \dots, n$, and $\sum_{i=1}^n p_i = 1$. Show that

$$n^3 + 2n + \frac{1}{n} \leq \sum_{i=1}^n \left(p_i + \frac{1}{p_i} \right)^2.$$

Hint: The Cauchy-Schwartz inequality can be helpful. It is useful to note two tricks:

$$\begin{aligned} \left(\sum_i a_i \right)^2 &\leq n \sum_i a_i^2, \\ \left(\sum_i a_i \right)^2 &\leq \left(\sum_i |a_i|^{2p} \right) \left(\sum_i |a_i|^{2(1-p)} \right). \end{aligned}$$

If you use these tricks make sure to justify them.

Problem 2. Let $a \in \mathbb{R}^n$. Let D be the $n \times n$ diagonal matrix with diagonal entries $D_{i,i} = (-1)^i$. Let $\lceil t \rceil$ denote the smallest integer bigger than or equal to t . Let $e \in \mathbb{R}^n$ such that $e_i = 1$ for $i = 1, \dots, n$. Show that

$$|e^T a|^2 + |e^T D a|^2 \leq 2 \left\lceil \frac{n}{2} \right\rceil \|a\|_2^2.$$

Problem 3.

- a) Let T be an $n \times n$ Toeplitz matrix. Let Z_n denote the $n \times n$ shift-up matrix. What is the maximum rank of the matrix $T Z_n - Z_n T$?
- b) Let V be an $n \times n$ unit upper-triangular Toeplitz matrix. Show that $Z_n = V Z_n V^{-1}$.
- c) Find all solutions X_R of the equation $X_R Z_n - Z_n X_R = 0$. What is the dimension of the solution space?
- d) Find all solutions Y_L of the equation $Y_L Z_n^T - Z_n^T Y_L = 0$. What is the dimension of the solution space?
- e) Consider the equation $X Z_n - Z_n X = Y$. Show that this equation has a solution X , if and only if, $\text{trace}(Y_L^T Y) = 0$, where Y_L is from part d). *Hint:* How many linearly independent constraints on Y does this give? How does that compare to the dimension of the null-space? What do you know about the angle between the range space and left null-space of a linear operator?
- f) Give an efficient algorithm to find a particular solution X_p such that $X_p Z_n - Z_n X_p = Y$, assuming Y is such that a solution exists.
- g) Find all solutions X of $X Z_n - Z_n X = Y$, assuming Y is such that a solution exists. *Hint:* You can express X as the sum of the particular solution X_p from part f) and a general homogenous solution.
- h) Given Y , show that the equation $X - Z_n^T X Z_n = Y$ always has a unique solution X and give an efficient algorithm to compute X from Y .